

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

UB E-Health: AI-Powered Integrated Healthcare & Hospital Management System

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- Conventional healthcare relies heavily on manual processes and paper-based records
- Patients face difficulty scheduling appointments, accessing reports, and understanding clinical terminology
- Doctors struggle with disorganised patient histories and inefficient prescription workflows
- Lab staff manage test requests manually — causing delays, lost samples, and entry errors
- No unified digital platform results in poor coordination between departments
- Increased administrative burden on staff and delayed access to medical information
- Unnecessary in-person visits for minor queries reduce quality of patient care
- Especially critical in resource-limited or offline healthcare environments

IN SCOPE

- Web-based app accessible via modern browsers
- Four roles: Patient, Doctor, Lab Personnel, Admin
- Appointment booking with online payment (Razorpay)
- Lab test management with home service option
- AI report interpretation (Gemini + Ollama fallback)
- Real-time doctor-patient messaging (Socket.io)
- Prescription management with PDF generation
- Health trend analytics & visualization
- Email notification system with cron scheduling
- Ambulance service management

OUT OF SCOPE

- Mobile native application (iOS / Android)
- Video consultation via WebRTC
- Integration with external hospital ERP systems
- Medical imaging analysis (X-ray, MRI interpretation)

MERN Stack Full-Stack Web Application

- Frontend: React.js :3000 → REST API (HTTP) + WebSocket
- Backend: Node.js + Express.js :3001
- Database Layer: MongoDB (data) + Redis Cache (presence / rate-limiting)
- Real-Time: Socket.io Server for doctor-patient chat
- **AI Services:**
 - Google Gemini API (Online cloud AI)
 - Ollama Local Model (Offline fallback)
 - Rule-Based Engine (Final fallback)
- Email: Gmail SMTP via Nodemailer
- Security: JWT Auth + bcrypt hashing + input sanitization + MIME validation

Sprint 1: Auth & Users

- Patient / Doctor / Admin / Lab registration
- JWT authentication middleware
- Role-based route protection

Sprint 2: Appointments & Lab

- Appointment booking with slot management
- Lab test booking + home service
- Razorpay integration + email notifications

Sprint 3: Records & Prescriptions

- Doctor report creation with vitals
- Prescription mgmt + PDF generation
- Document upload & sharing

Sprint 4: AI Integration

- Google Gemini API for report interpretation
- Ollama local AI fallback
- Rule-based fallback + health chatbot

Sprint 5: Real-Time Chat

- Socket.io server setup
- Redis for caching & presence
- Doctor-patient chat + typing indicators

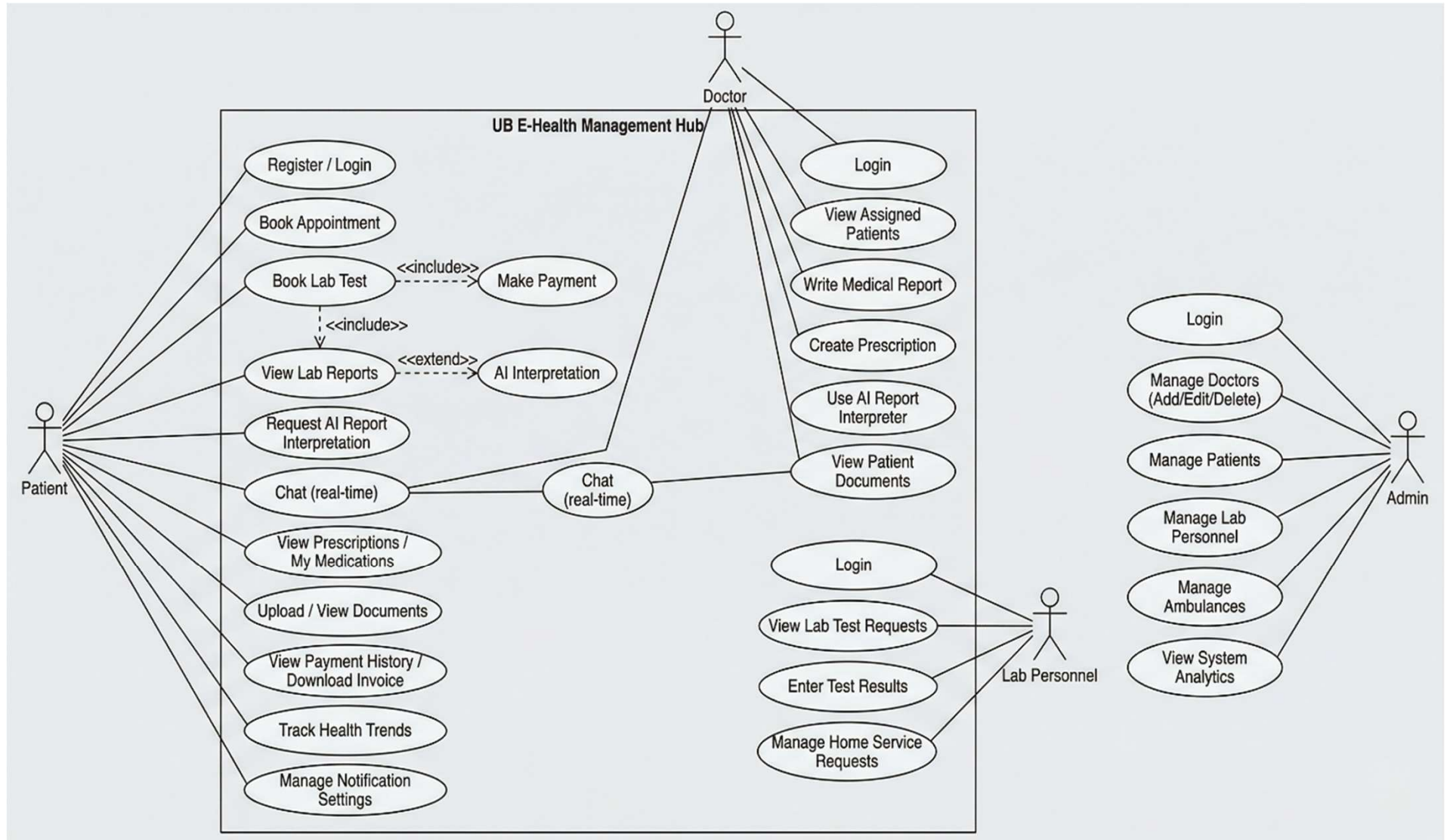
Sprint 6: Analytics & Polish

- Health trends analytics dashboard
- Health score calculation system
- Admin dashboard + responsive UI

- React.js v18 — Component-based UI; virtual DOM for efficient rendering
- Redux — Centralized state management; unidirectional data flow
- React Router v6 — Client-side routing with protected routes for role auth
- Axios — Promise-based HTTP client with request/response interceptors
- Chart.js + react-chartjs-2 — Interactive health trend charts (vitals over time)
- jsPDF — Client-side PDF generation for prescriptions and appointment receipts
- React-Toastify — Non-intrusive success / error / info notifications
- socket.io-client — WebSocket connections for real-time doctor-patient chat

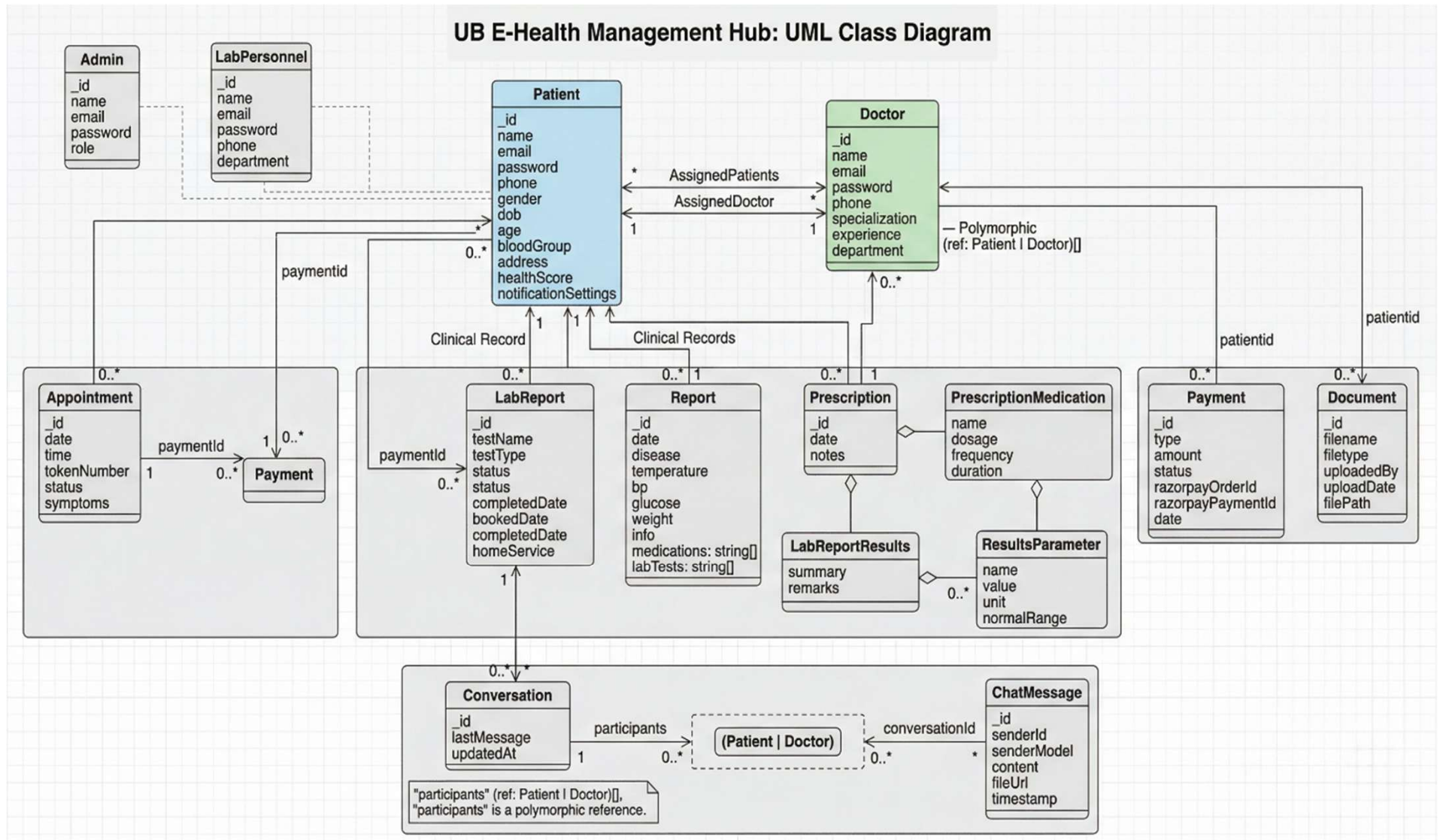
- Node.js — Non-blocking event-driven JavaScript runtime; ideal for real-time I/O
- Express.js v4 — REST API framework with middleware and routing
- Socket.io v4 — Bidirectional real-time communication with WebSocket + HTTP fallback
- JWT (jsonwebtoken) — Stateless authentication; token carries userId and role
- bcrypt — Password hashing with salt; prevents rainbow table attacks
- Multer — Middleware for multipart file uploads (medical docs, chat attachments)
- MongoDB + Mongoose v6 — NoSQL document DB with ODM schema, validation & populate()
- Redis (ioredis) — In-memory store for presence, typing indicators, rate limiting
- Nodemailer + node-cron — Automated appointment / medication email reminders
- Winston — Audit logging for chat, auth failures, and security events

System Diagrams — Use Case Diagram



Four actor roles: Patient, Doctor, Lab Personnel, Admin — with AI report interpretation and real-time chat interactions

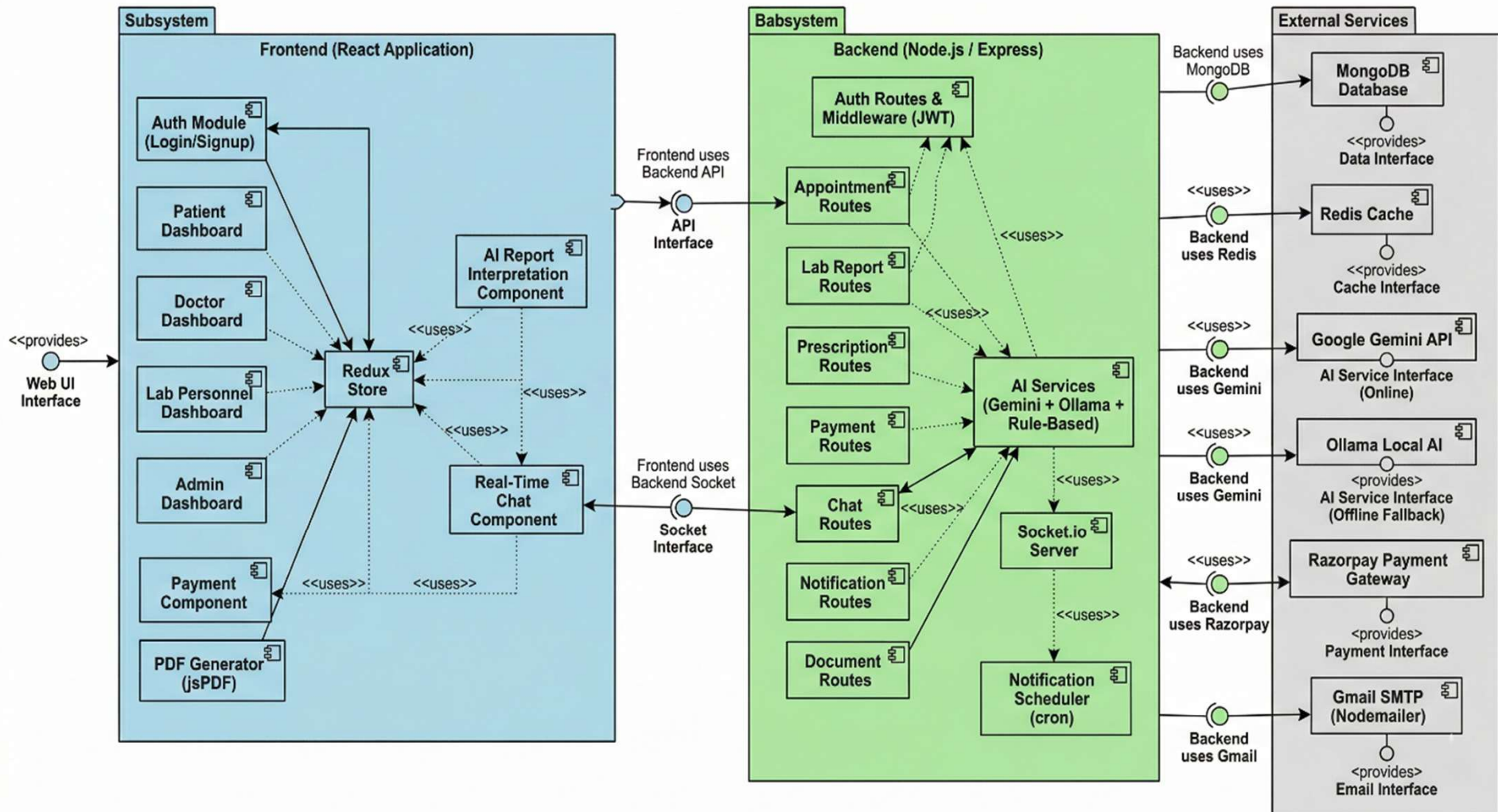
System Diagrams — UML Class Diagram



Core entities: Patient, Doctor, LabPersonnel, Admin with relationships to Appointment, LabReport, Prescription, Conversation, ChatMessage, Payment, Document

System Diagrams — UML Component Diagram

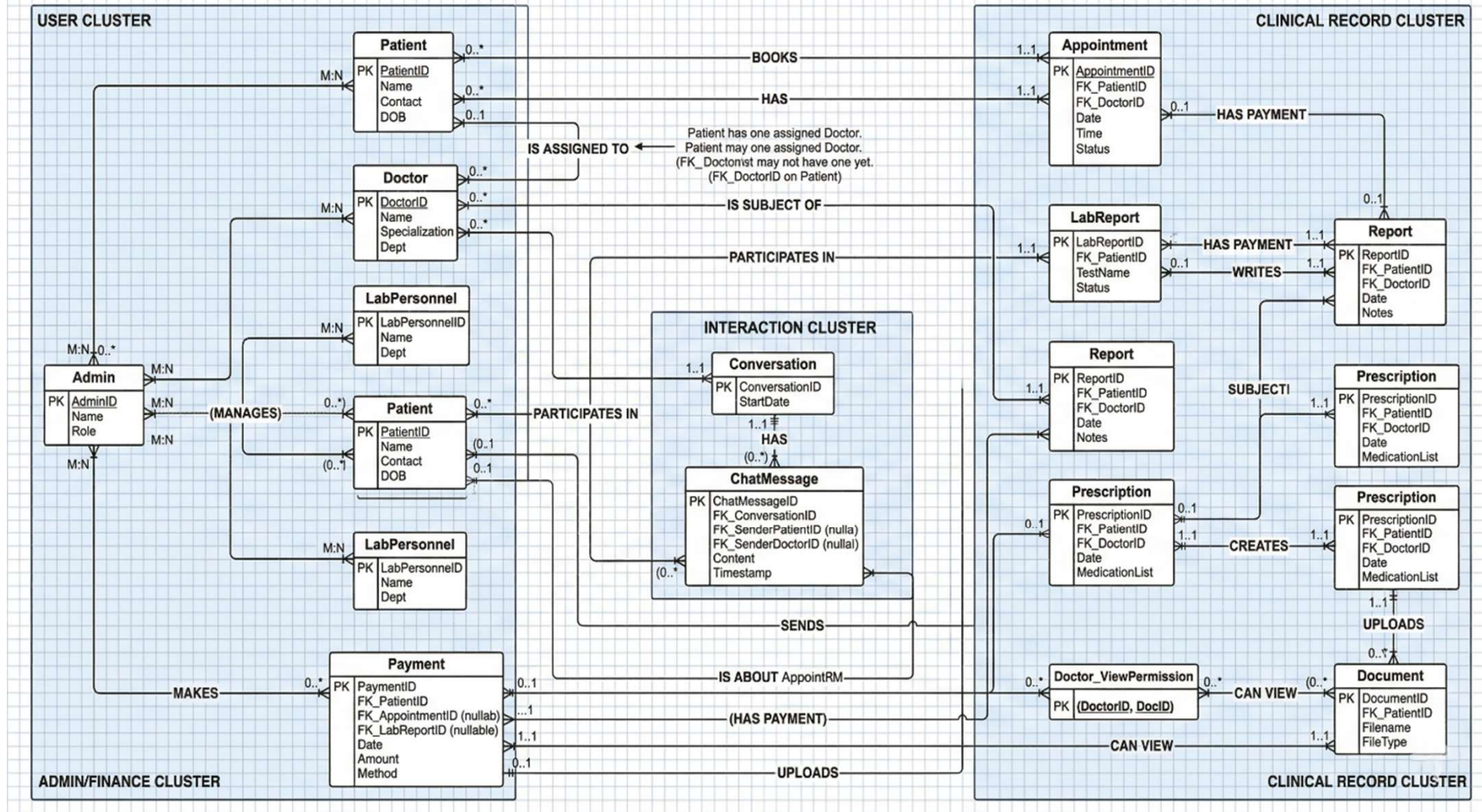
UB E-Health Management Hub: UML Component Diagram



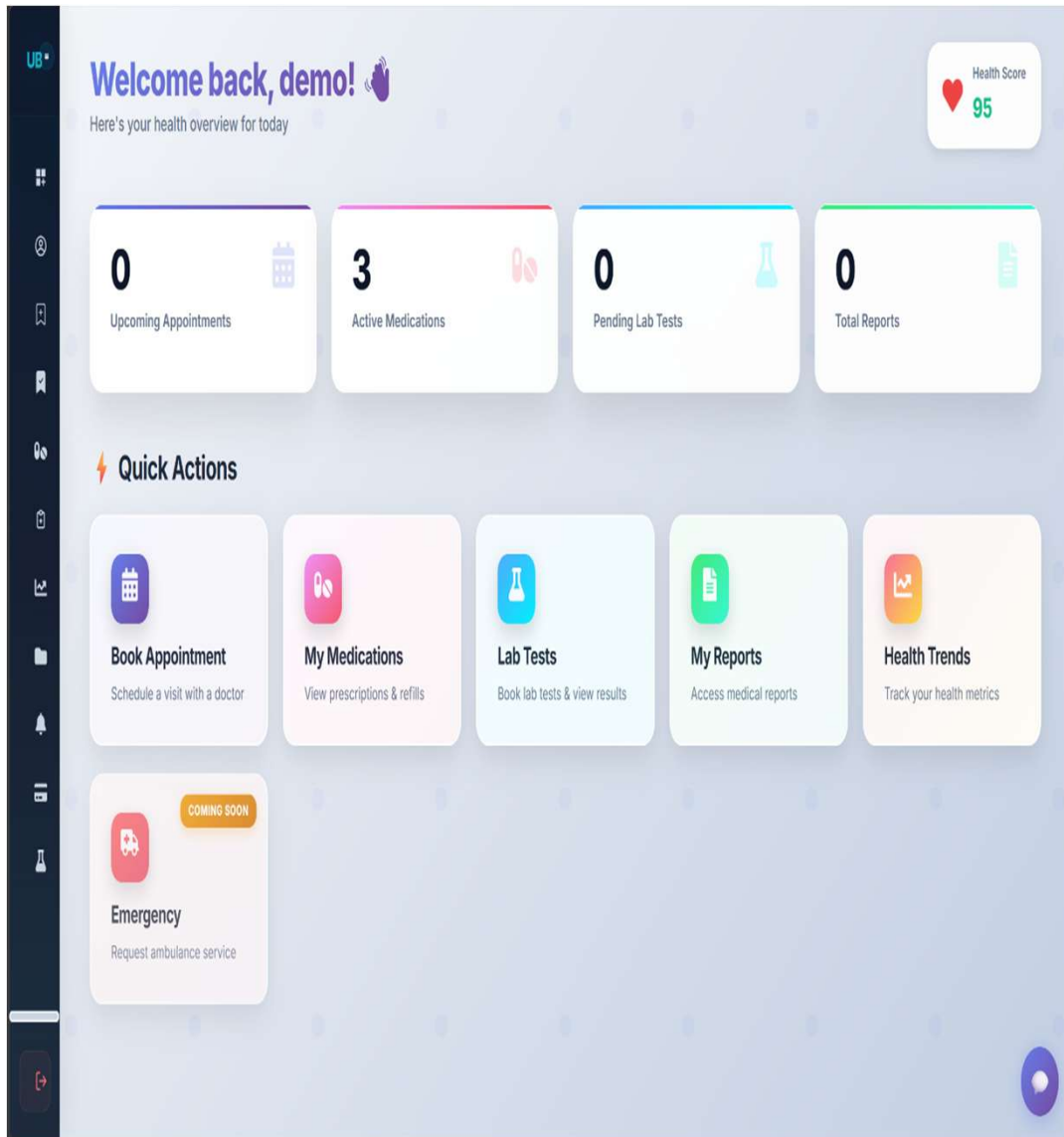
Frontend (React + Redux Store) communicates with Backend (Express routes, AI services, Socket.io, Notification Scheduler) and External Services (MongoDB, Redis, Gemini, Razorpay, Gmail)

System Diagrams — Database ER Diagram

UB E-Health Management Hub: Database ER Diagram (Crow's Foot Notation)



Crow's Foot Notation: User Cluster (Patient, Doctor, LabPersonnel, Admin), Clinical Record Cluster (Appointment, LabReport, Report, Prescription), Interaction Cluster (Conversation, ChatMessage)



Patient Dashboard

- Health Score indicator (0-100)
- Stat cards: Upcoming Appointments, Active Medications, Pending Lab Tests, Total Reports
- Quick Action tiles: Book Appointment, My Medications, Lab Tests, My Reports, Health Trends
- Emergency (Ambulance) service — Coming Soon

Application Output — Appointment Booking & AI Chatbot

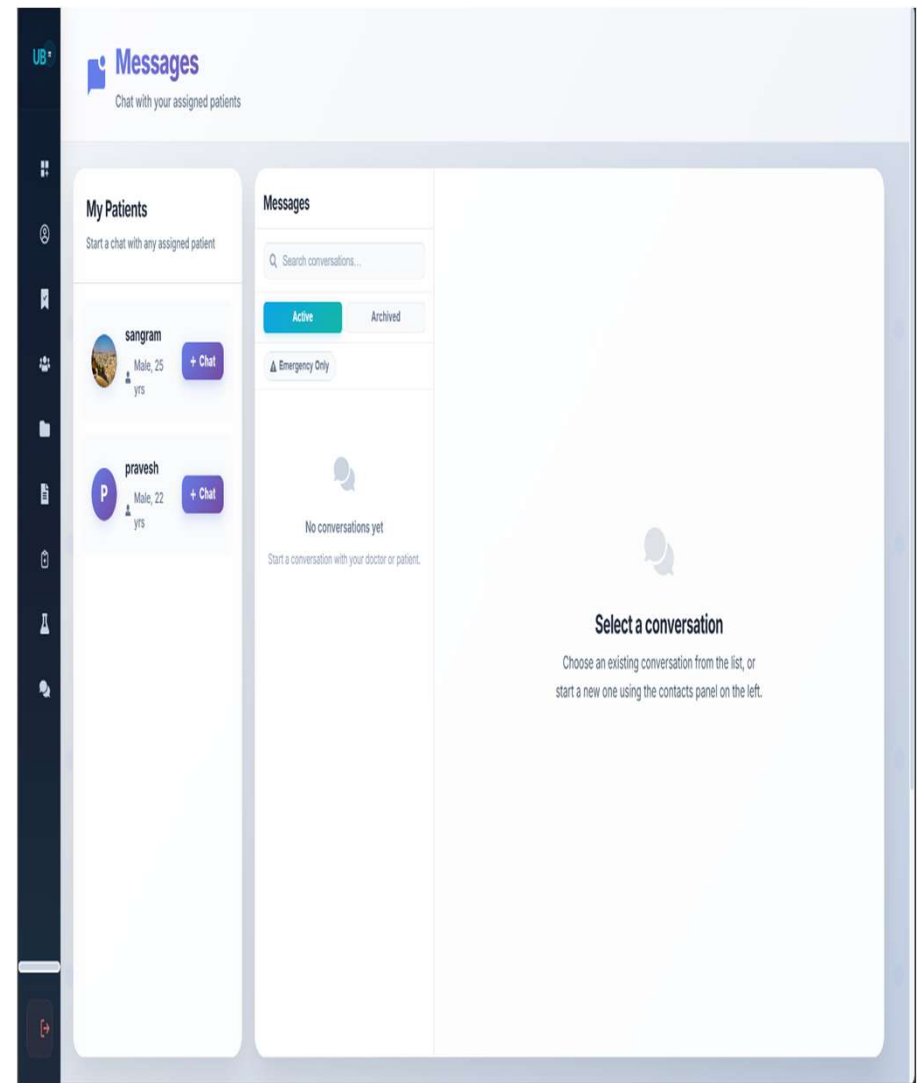
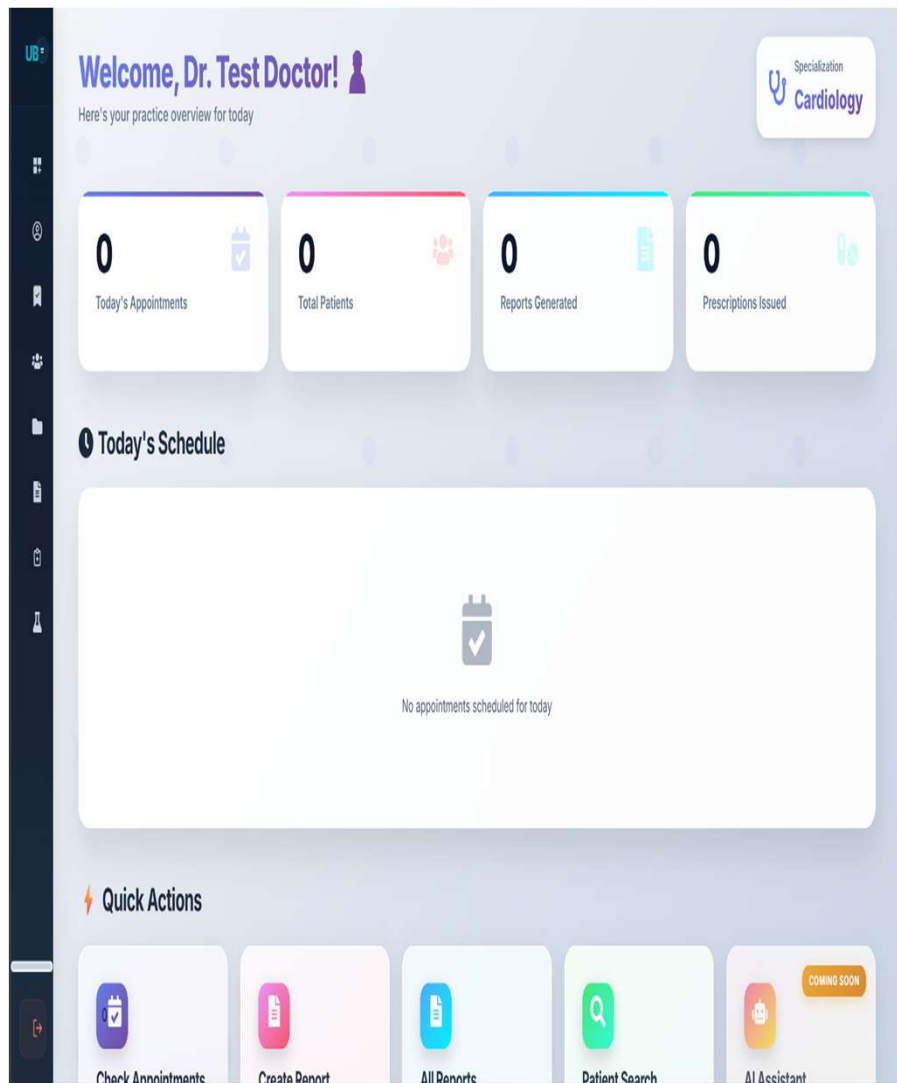
The screenshot shows the 'Book Appointment' form. At the top, there are two tabs: 'Doctor Appointment' (active) and 'Lab Test'. The form is titled 'Book Appointment' with the subtitle 'Select Department & Doctor'. It contains several input fields: 'Department' (a dropdown menu), 'Doctor' (a dropdown menu), 'Fees' (a text input field), 'Problem' (a text input field with the placeholder 'Describe your issue'), and 'Date & Time' (which includes a date picker showing 'dd-mm-yyyy' and a time dropdown menu). At the bottom of the form is a green button labeled 'Proceed to Payment'.

The screenshot shows the 'Patient Profile' page. At the top, there is a header 'Patient Profile' with the subtitle 'Manage your personal information and account settings'. The page is divided into several sections: a profile card on the left with a user avatar, name 'demo', patient ID 'd0a3b4d1', phone number '1234567890', email 'sk.shalishav.13@gmail.com', and date of birth '10/12/2003'; a 'Personal Information' section with fields for 'Gender' (Male), 'Age' (22 years old), 'Blood Type' (AB+), and 'Address' (123, abc); a 'Hospital Information' section with fields for 'Hospital Hours' (09:00 AM - 08:00 PM), 'Hospital Name' (UBHospital), and 'Hospital Address' (123, ABC); and a 'Health Assistant' section with a '24/7 Emergency' button. At the bottom, there are buttons for 'Password', 'Edit', and 'Activity History'.

Appointment Booking Form: Select Department, Doctor, Date & Time, then proceed to Razorpay payment

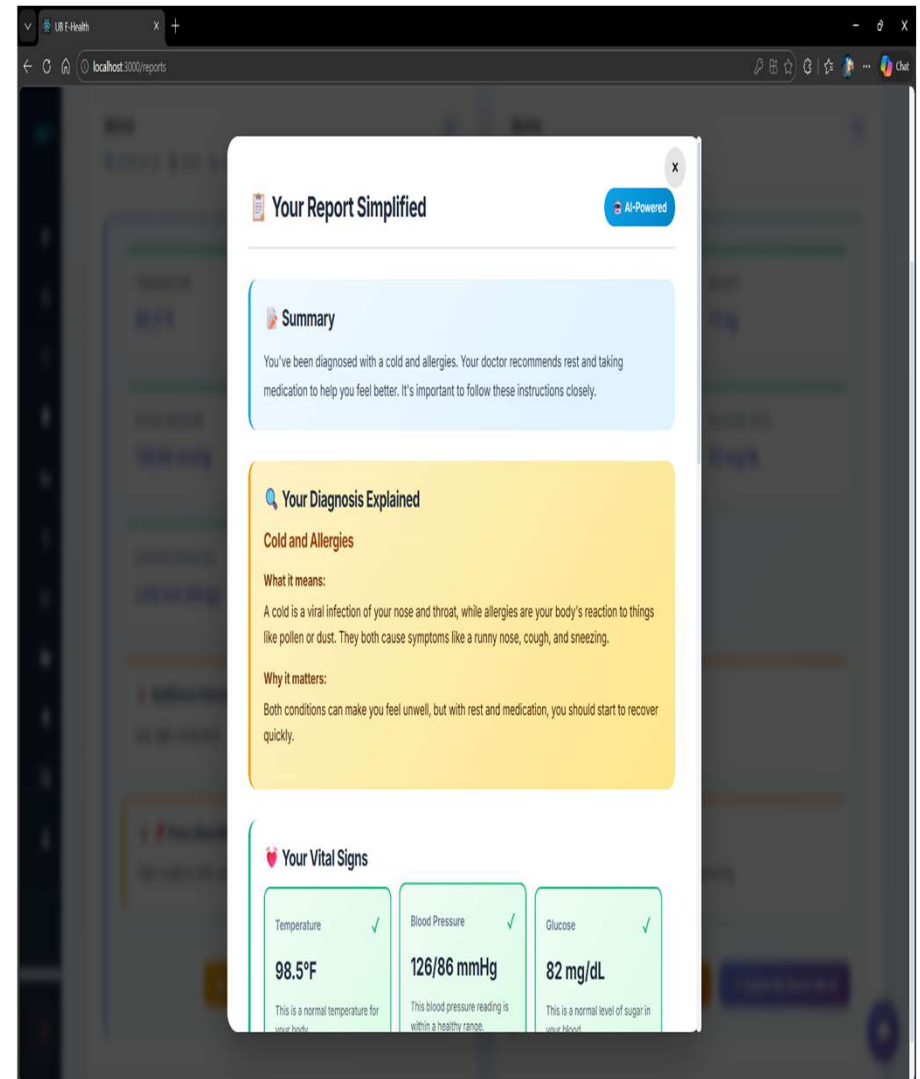
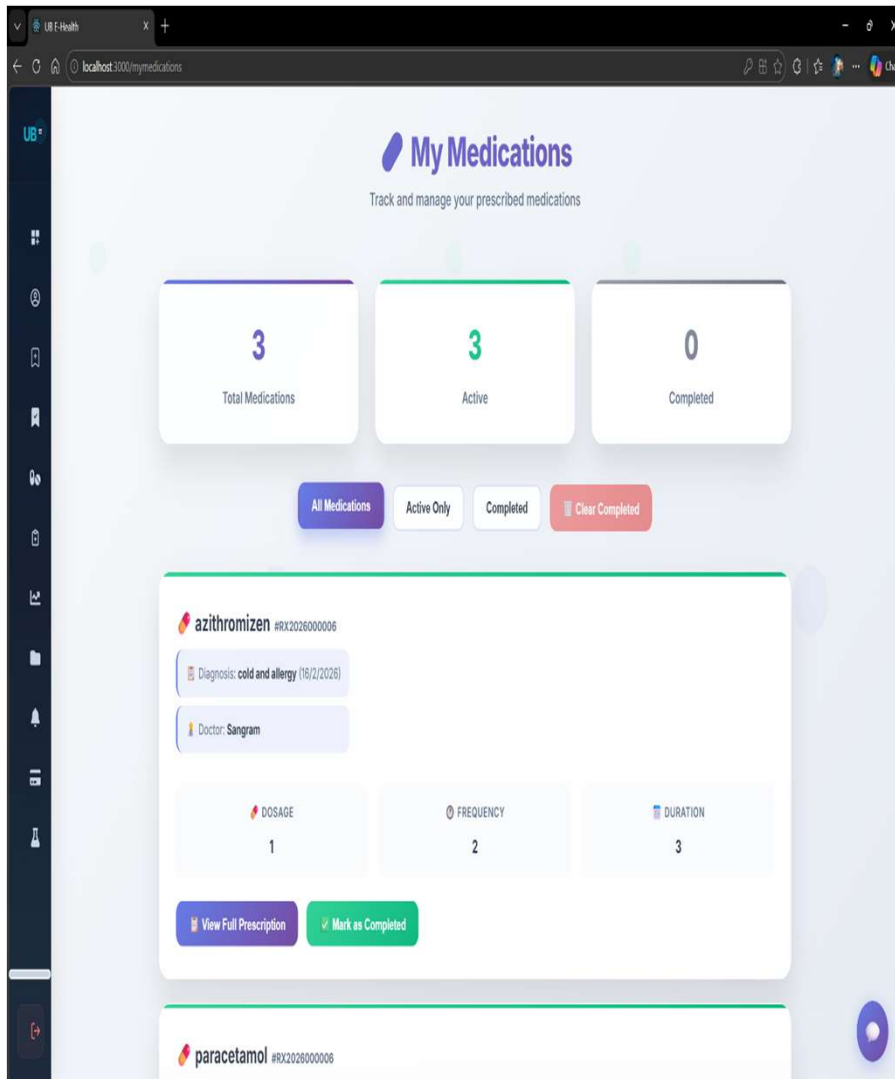
Patient Profile + AI Health Chatbot: Personal info, Health Grade, Hospital details, and 24/7 AI assistant

Application Output — Doctor Dashboard & Messaging



Doctor Dashboard: Today's Appointments, Total Patients, Reports Generated, Prescriptions Issued — with Quick Actions
Real-Time Messaging: Doctor-Patient chat via Socket.io; My Patients list with active conversations and emergency-only mode

Application Output — Medications & AI Report Interpreter



My Medications: Track active prescriptions — Dosage, Frequency, Duration — with View Full Prescription and Mark as Completed

AI Report Simplified: Executive Summary, Diagnosis Explained, Vital Signs — all in plain patient-friendly language powered by Google Gemini

AI Feature Outcomes

- Patient Comprehension: Lab reports translated to plain language without medical training
- Offline Availability: Ollama fallback keeps AI functional in rural/offline settings
- Structured Output: JSON response in sections — Summary, Parameter Explanations, Recommendations, Doctor Questions
- Safety Disclaimer: Every interpretation includes a reminder to consult their doctor

Security Outcomes

- XSS Prevention: All user inputs sanitized; script tags stripped via DOMPurify
- Injection Prevention: ObjectId validation prevents NoSQL injection attacks
- Unauthorized Access: Role middleware blocks all cross-role access attempts
- Password Security: bcrypt hashing with salt rounds for all stored credentials
- File Upload Security: MIME type and file size validation enforced on every upload

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Conclusion

- Successfully built a production-ready AI-powered healthcare management system on the MERN stack
- All planned objectives achieved — role auth, appointments, lab tests, prescriptions, AI reports, real-time chat
- Dual-mode AI architecture (Gemini + Ollama) — a key original contribution making intelligent interpretation universally accessible
- Security implemented at every layer; system tested end-to-end across all modules
- Reduces appointment scheduling time, improves patient understanding, centralises hospital workflows

Future Scope

- WebRTC video consultation (telemedicine)
- React Native mobile app for iOS & Android
- Medical image analysis with vision-capable AI (X-ray, MRI, CT)
- Wearable device integration (smartwatch health metrics)
- Two-Factor Authentication (OTP via SMS)
- Multi-language support for regional Indian languages
- HL7 FHIR-compliant EHR export for interoperability
- HIPAA/DPDP-compliant audit trail and compliance reporting

Thank You



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Tech Stack: MERN | Socket.io | Redis | Google Gemini | Ollama | Razorpay | JWT